



Service Tool Instruction

Service Tool Instruction
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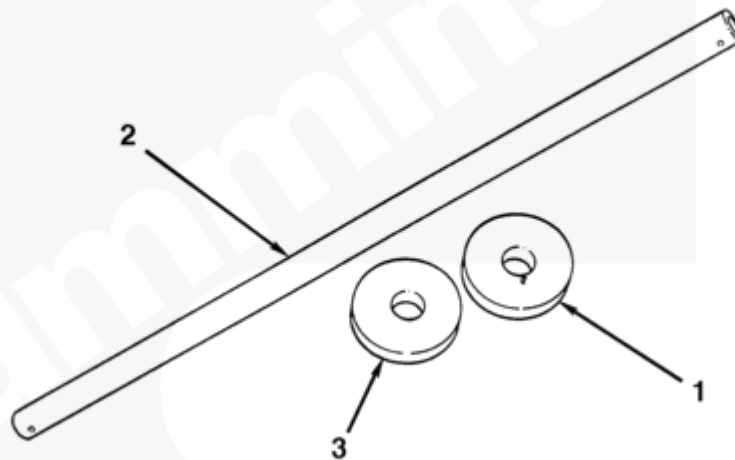
Description

Main Bearing Alignment Kit

Purpose

This service tool instruction provides information on the main bearing alignment kits. The main bearing alignment kit contains the necessary tools to check the main bearing bore alignment of various cylinder blocks. The main bearing alignment kits are available for the NH, NT, N14, V28, K19, K38, K50, V903, V504, V555, C, J, L10, M11, ISM, C8.3, QST30, QSK45, and QSK60 engines.

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Table 1, Main Bearing Alignment Kit (NH, NT, and N14), Part Number 3375847

Item Number	Part Number	Description	Quantity
1	ST 1177-39	Centering ring	2
2	ST 1177-16	Checking bar	1
3	ST 1177-13	Checking ring	1

Table 2, Main Bearing Alignment Kit (V12 phase I and II), Part Number 3375848

Item Number	Part Number	Description	Quantity
1	ST 1177-41	Centering ring	3

Table 2, Main Bearing Alignment Kit (V12 phase I and II), Part Number 3375848			
Item Number	Part Number	Description	Quantity
2	ST 1177-16	Checking bar	1
3	ST 1177-14	Checking ring	1
Table 3, Main Bearing Alignment Kit (V28 phase III), Part Number 3375849			
Item Number	Part Number	Description	Quantity
1	ST 1177-42	Centering ring	3
2	ST 1177-16	Checking bar	1
3	ST 1177-15	Checking ring	1
Table 4, Main Bearing Alignment Kit (K19), Part Number 3375850			
Item Number	Part Number	Description	Quantity
1	ST 1177-54	Centering ring	2
2	ST 1177-16	Checking bar	1
3	ST 1177-55	Checking ring	1
Table 5, Main Bearing Alignment Kit (K38 and K50), Part Number 3375851			
Item Number	Part Number	Description	Quantity
1	3375063	Centering ring	3
2	3375061	Checking bar	1
3	3375064	Checking ring	1
Table 6, Main Bearing Alignment Kit (V903), Part Number 3375852			
Item Number	Part Number	Description	Quantity
1	ST 1177-37	Centering ring	2
2	ST 1177-16	Checking bar	1
3	ST 1177-11	Checking ring	1
Table 7, Main Bearing Alignment Kit (V504 and V555), Part Number 3375853			
Item Number	Part Number	Description	Quantity
1	ST 1177-39	Centering ring	2
2	ST 1177-16	Checking bar	1
3	ST 1177-9	Checking ring	1

Table 8, Main Bearing Alignment Kit (C and J), Part Number 3375854

Item Number	Part Number	Description	Quantity
1	ST 1177-38	Centering ring	2
2	ST 1177-16	Checking bar	1
3	ST 1177-12	Checking ring	1

Table 9, Main Bearing Alignment Kit (L10, M11, and ISM), Part Number 3822497

Item Number	Part Number	Description	Quantity
1	3376781	Centering ring	2
2	ST 1177-16	Checking bar	1
3	3376811	Checking ring	1

Table 10, Main Bearing Alignment Kit (C8.3), Part Number 3824579

Item Number	Part Number	Description	Quantity
1	3824581	Centering ring	2
2	ST 1177-16	Checking bar	1
3	3824580	Checking ring	1

Table 11, Main Bearing Alignment Kit (QST30), Part Number 3164758 *

Item Number	Part Number	Description	Quantity
1	3164760	Centering ring	3
3	3164759	Checking ring	1

Table 12, Main Bearing Alignment Kit (QSK45 and QSK60), Part Number 3164874 **

Item Number	Part Number	Description	Quantity
1	3164873	Centering ring	3
3	3164872	Checking ring	1

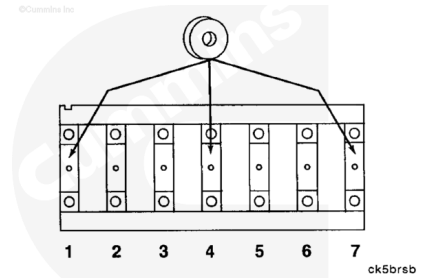
Note : * Part Number ST 1177-16 checking bar is required for use with these checking ring and centering rings and must be ordered separately.

Note : ** Part Number 3375061 checking bar is required for use with these checking ring and centering rings and must be ordered separately.

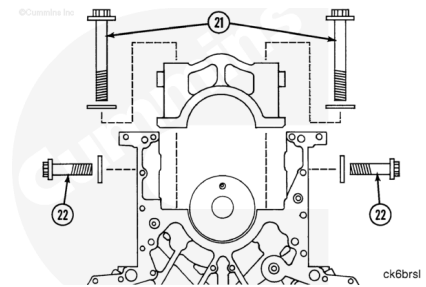
Note : The illustrations show checking the main bearing bore on a V28 series engine. The process is similar on all other engine families.

Allow the checking bar, centering rings, checking ring and the cylinder block to stabilize to room temperature.

Install the centering rings. Centering ring location will vary depending on engine model. See the Shop Manual for detailed instructions.



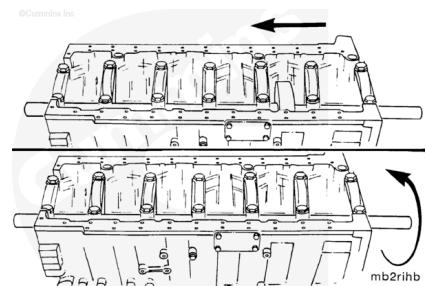
Install the main bearing caps, refer to the appropriate Shop Manual for the installation procedures and specifications.



Use clean 15W-40 oil to lubricate the checking ring, checking bar and centering rings.

Slide the checking bar through one of the centering rings and install the checking ring on the checking bar. Continue pushing the checking bar through the rest of the centering rings.

Inspect the checking bar for freedom of rotation.

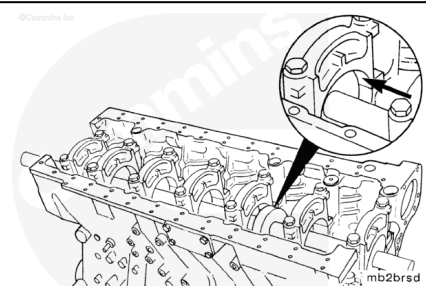


Push the checking ring through the main bearing bores to be inspected using light finger pressure on each side of the checking ring.

The checking ring **must** pass through each bore with slight resistance.

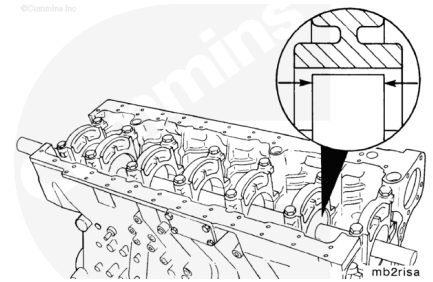
If the checking ring will **not** go through the bore, inspect the bore for burrs and other obstructions. Remove any burrs if necessary.

If the checking ring does **not** pass through the main bearing bore, an undersized condition exists and the cylinder block bore **must** be repaired.

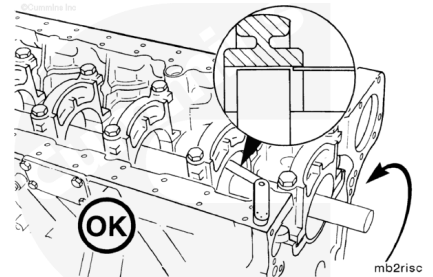


If the checking ring will pass through the main bearing bore without any resistance, center the checking ring in the bore. Use a feeler gauge 0.08 mm [0.003 inch] that is **not** more than

12.7 mm [1/2 inch] wide. Insert the feeler gauge between the checking ring and the cylinder block.



Evaluate the main bearing bore using the feeler gauge. Refer to the appropriate Shop Manual for the detailed inspection procedures.



Remove the checking ring, checking bar and centering rings.

